

Spring-18

- * dynamic programming vs greedy algo.
- * Huffman tree.
- * Activity selection problem - find all optimal set. $\longleftrightarrow$ activity selection problem.
- * DFS.
- * Adjacency matrix.
- * Show topological sort (unweighted graph).
- * Shortest path. $\odot$
- * Bellman-ford's shortest path.
- * Neg-weight cycle.
- * Backtracking (N-Rook pbl).
- * Define P, NP / NP-complete.
- * Relaxation vertex.
- * $x^n \bmod m$
- * point/vertex using Graham's scan algo.
- * line segment properties.

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- * Travelling salesman problem
- * Optimal Huffman code
- * Activity selection problem ~~***~~
- * DFS
- * Spanning tree / minimum spanning tree (Kruskal Algo) ~~***~~
- * Visit by DFS + BFS
- * Shortest path
- * Dijkstra Algo.
- * relaxation of vertex
- * N-P complete
- * Graham's scan algo.
- * $u^n \bmod m$ ~~***~~
- * Branch / Bound

* greedy choice / knapsack prob.

* Huffman tree

* shortest path

* BFS / DFS.

* Kruskal Algo - minimum spanning tree

* minimum spanning tree / Prim algo.

* Kruskal $\leftrightarrow$ MST

* Invariant s. path algo.

* Relaxation of a vertex

* Dijkstra algo $\rightarrow$ (s. path)

* Backtracking (i-Rock).

* P/NP/ NP-complete

* Convex hull prob.

* Graham scan algo.

* Floyd-Algo.

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greedy stg $\leftrightarrow$ 0/1 knapsack

Huffman coding

BFS.

DFS.

Adjacency list

Spanning tree $\leftrightarrow$ Prim algo

Kruskal Algo.

Shortest path.

Bellman-ford algo.

Floyd-Warshall

Two line segment collinear or not

Subpaths are shortest path

P/NP/ NP-complete

N-queen problem

Branch and bound algo $\rightarrow$ solve $\rightarrow$ it travelling salesman problem

Cross product

greedy choice

greedy algo.

Huffman code

Prim algo.

Dijkstra

code $\leftrightarrow$ Un-... direct graph

m. Spanning tree

MST for weighted graph

Dijkstra's algo $\leftrightarrow$ single source sh... path

Bellman-ford algo

Sho... path $\leftrightarrow$ Floyd ~~war~~ Warshall algo

Cross product

Graham's scan algo

NP-complete

branch/Bound

$\leftarrow$ Rock problem (Backtracking).